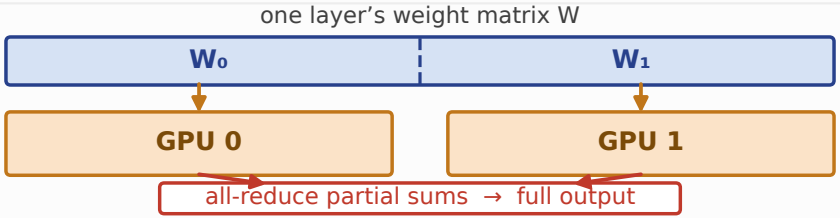
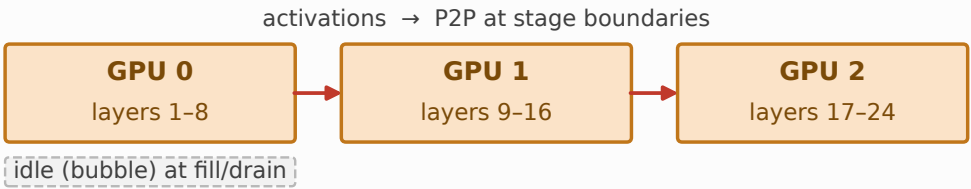
A · Tensor parallel (TP) — splits one layer's weights



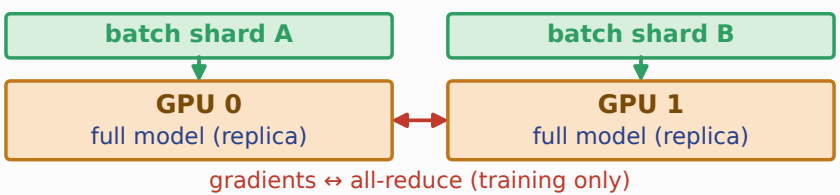
communication: all-reduce every layer → on-node only (NVLink)

B · Pipeline parallel (PP) — splits the layer stack



communication-light: only stage boundaries move → spans nodes

C · Data parallel (DP) — splits the data batch



inference: independent replicas, no sync · throughput scales linearly

blue = weights · orange = GPU · green = data · red = comms